

SMITHTON, Australia

WMO: 949530

Lat: 40.8347S

Lon: 145.0847E

Elev: 29

StdP: 14.68

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.8	34.7	31.5	25.9	35.6	33.1	27.7	37.5	30.2	51.1	27.5	50.5	5.5	190	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	16.3	75.4	65.6	72.9	64.3	70.8	63.0	67.4	72.9	65.9	70.8	64.5	69.1	15.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.4	94.0	69.7	63.8	88.8	68.8	62.4	84.5	67.8	31.8	73.2	30.7	70.8	29.7	69.1	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.6	25.8	23.4	DB	29.2	82.1	1.5	4.7	28.1	85.4	27.3	88.2	26.4	90.8	25.3	94.2
			WB	29.2	70.6	1.5	1.9	28.1	72.0	27.3	73.2	26.5	74.3	25.4	75.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	60.5	61.2	58.5	54.4	51.2	48.5	47.9	48.2	49.9	51.9	55.3	57.5	(6)
	DBStd	6.34	4.83	4.93	5.03	4.70	4.28	4.20	3.70	3.74	3.69	4.08	4.47	4.47	(7)
	HDD50	374	0	0	2	12	37	78	87	80	47	25	5	1	(8)
	HDD65	4173	157	126	212	319	428	494	530	520	454	405	291	237	(9)
	CDD50	1728	325	313	264	144	74	34	22	25	43	85	165	234	(10)
	CDD65	50	16	19	9	0	0	0	0	0	0	0	1	5	(11)
	CDH74	158	74	42	24	1	0	0	0	0	0	0	3	14	(12)
	CDH80	17	12	1	3	0	0	0	0	0	0	0	0	1	(13)
(14) Wind	WSAvg	11.9	13.8	13.2	11.8	10.4	10.2	9.6	10.6	11.2	12.2	12.6	13.0	13.8	(14)
(15) Precipitation	PrecAvg	43.5	2.0	1.8	2.4	3.9	4.6	4.5	5.6	5.1	4.5	3.5	3.0	2.8	(15)
	PrecMax	57.2	4.7	6.1	6.2	9.5	9.8	8.4	9.8	10.4	8.3	7.5	6.0	9.0	(16)
	PrecMin	31.5	0.5	0.2	0.3	0.8	1.0	1.7	2.5	1.2	0.7	1.2	1.2	0.0	(17)
	PrecStd	7.2	1.0	1.3	1.5	2.0	2.3	1.5	1.9	2.1	1.9	1.6	1.2	1.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.3	78.4	76.9	69.6	63.9	60.1	58.7	60.2	63.0	68.0	72.6	76.2	(19)
		MCWB	67.9	67.0	67.0	62.4	58.5	55.9	54.6	54.4	56.3	59.1	63.0	65.8	(20)
	2%	DB	75.3	74.9	72.7	66.8	62.0	58.4	56.8	58.0	60.3	64.0	69.0	71.8	(21)
		MCWB	65.4	65.8	64.8	61.1	57.9	55.0	52.9	53.2	54.8	57.1	61.3	62.4	(22)
	5%	DB	72.4	72.5	70.0	64.9	60.5	57.1	55.6	56.5	58.5	61.8	66.3	68.9	(23)
		MCWB	63.8	64.4	63.3	60.0	57.0	54.0	52.0	52.3	53.6	56.1	59.5	61.2	(24)
	10%	DB	69.9	70.4	67.9	63.4	59.2	56.0	54.6	55.1	57.1	59.9	63.9	66.6	(25)
		MCWB	62.1	63.0	61.7	59.0	56.0	53.1	51.3	51.5	52.7	54.6	57.8	60.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	69.1	68.9	64.5	60.2	57.5	55.6	55.8	57.5	60.9	65.1	67.0	(27)
		MCDB	76.8	74.8	73.8	67.3	62.2	59.1	57.4	58.4	60.8	64.2	69.6	72.5	(28)
	2%	WB	66.9	67.4	66.2	62.5	58.9	55.8	53.9	54.1	56.0	58.8	62.6	64.6	(29)
		MCDB	73.2	72.6	70.8	65.1	61.0	57.5	55.8	56.7	58.9	62.8	67.7	69.0	(30)
	5%	WB	65.2	66.0	64.4	61.2	57.6	54.7	52.8	53.1	54.7	57.2	60.5	62.6	(31)
		MCDB	70.4	70.6	68.7	64.3	59.9	56.6	54.9	55.7	57.5	60.7	64.8	67.0	(32)
	10%	WB	63.4	64.6	62.7	59.6	56.4	53.5	51.8	52.1	53.5	55.6	58.7	60.9	(33)
		MCDB	68.3	69.2	66.5	62.8	58.8	55.7	54.1	54.7	56.2	59.2	63.1	65.5	(34)
(35) Mean Daily Temperature Range		MDBR	17.0	16.3	16.5	15.8	14.0	13.4	12.6	13.1	13.4	15.1	15.1	15.9	(35)
	5% DB	MCDBR	20.0	17.6	18.0	16.7	14.2	12.7	12.5	13.5	14.8	16.1	17.4	18.8	(36)
		MCWBR	11.4	10.2	11.1	11.6	10.8	10.0	9.4	9.9	10.2	10.7	10.7	11.3	(37)
	5% WB	MCDBR	16.5	14.0	15.1	13.7	12.6	11.4	11.2	12.4	12.9	14.3	14.6	15.5	(38)
		MCWBR	10.7	9.3	10.5	10.5	10.4	9.6	9.2	9.9	9.9	10.6	9.7	10.4	(39)
(40) Clear-Sky Solar Irradiance	taub	0.352	0.343	0.333	0.331	0.318	0.311	0.311	0.311	0.331	0.366	0.344	0.345	0.342	(40)
	taud	2.504	2.547	2.576	2.554	2.579	2.585	2.589	2.524	2.396	2.488	2.503	2.529		(41)
	Ebn at Noon	310	305	294	272	255	246	255	268	277	301	310	315		(42)
	Edh at Noon	35	32	29	25	21	19	21	26	34	34	35	35		(43)
(44) All-Sky Solar Radiation	RadAvg	2119	1848	1386	946	607	484	528	781	1142	1618	1944	2112		(44)
	RadStd	134	92	67	58	41	40	31	60	70	92	118	142		(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) +1.06	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (16 neighbors)	+0.50	N/A	+0.54	+1.19	+0.72	+0.77	N/A	-155	+164	+18	(47)

Nomenclature: See separate page